

AI & Data Engineer focused on distributed systems, backend infrastructure, and enterprise AI platforms with hands-on experience building scalable execution, data processing, and observability systems.

ramaadithya5@gmail.com | +91 9940674086 | [linkedin.com/in/aadiithyyya/](https://www.linkedin.com/in/aadiithyyya/) | aadithyaram.vercel.app | github.com/aadiithyyya

EDUCATION

M.Tech (Integrated) Computer Science Engineering & Data Science Specialization | GPA: 9.57
Sep 2021 - June 2026 : SRM Institute of Science and Technology, Chennai

WORK EXPERIENCE

SAP Labs - AI & Data Engineering Intern
SAP BTP | Cloud Integration & Data Engineering Team

Aug 2025 - June 2026
Bangalore, KA

- Engineered MCP integration for observability, data pipelines, optimizing request routing, caching, and context propagation; **reducing token usage by ~78%** and **inference costs by ~57%**.
- Designed multi-agent execution system using LangGraph and FastAPI for data ingestion and **SRE incident analysis** across SAP BTP systems processing **~5B records/week** serving critical downstream applications.
- Resolved production scheduling, resource contention, and telemetry pipeline failures across distributed Kubernetes infrastructure serving **~70% of Fortune-500 customers**.
- Developed Databricks compute and cron-based alert systems across bronze, silver, and gold layers in lakehouse storage with automated anomaly detection and monitoring, **reducing outage MTTR by 35%**.

PROJECTS

Code Execution Runtime for MCP Systems
LangChain, LangGraph, MCP, FastAPI, PostgreSQL, Python

Mar 2026

- Loaded query-relevant MCP tool definitions dynamically, **reducing** tool-definition **context overhead by over 88%**.
- Executed selective tool invocation across 100+ mapped MCP tools while limiting active context to 1–4 definitions.
- Reduced inference latency, token overhead by filtering large tool outputs before model execution, **lowering final LLM input by 67%**.

Enterprise Supply Chain Intelligence & Forecasting Platform
Python, Node.Js, JavaScript, TensorFlow, PyTorch, GraphSAGE, Firebase, React, SQL

May 2025

- Designed batch inference systems for procurement forecasting using ensemble and LSTM prediction models.
- Reduced routing overhead through optimized **OSRM** path computation, **lowering** multi-stop **transit time by ~22%**.
- Integrated REST-based shipment coordination, supplier synchronization services across **10+ operational analysis dashboards**.

TECHNICAL EXPERTISE

- Languages:** Python, SQL
- Backend & Data Systems:** FastAPI, REST APIs, Node.js, PostgreSQL, SAP HANA
- Distributed Infrastructure:** Docker, Kubernetes, Databricks, CI/CD, Spark, Kafka, SAP BTP
- Cloud & Platforms:** SAP HANA Cloud, Cloud Foundry, AWS
- AI/ML Systems:** RAG, LangGraph, MCP, TensorFlow, PyTorch, MLOps, n8n, SAP AI Core
- Developer Tools:** Git, Jira, Firebase, Postman, VS Code

ACHIEVEMENTS

- SAP Recognition Award 2026** — Recognized for PoCs and productization of enterprise AI contributions.
- Volkswagen Hackathon 2025** — Top 50 finalist among 60K+ participants; earned PPO for OEM-infra solution.
- Citibank Hackathon 2024** — 4th Place Winner for AI-integrated financial systems relationship manager solution.